

Impact of the phase change material (PCM) integration on the mortar's thermal inertia: The role of the aggregates and PCM types

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ABSTRACT

Integrating phase change materials (PCMs) into building materials has gained attention for enhancing thermal performance and energy efficiency. However, the impact of different PCM types on cementitious materials' thermal and mechanical properties requires further study. This work addresses this gap by incorporating two bio-based esters and a paraffin-based PCM into natural aggregate mortar (NAM), recycled concrete aggregate mortar (RCAM), and lightweight aggregate mortar (LWAM). Thermal and mechanical properties were evaluated using the transient plane source method, heating and cooling cyclic tests, and standard mechanical tests. PCM impregnation was influenced by both PCM viscosity and mortar porosity. High-viscosity PCMs (paraffin) achieved greater absorption in more porous mortars like LWAM, while the incorporation of low-viscosity PCMs (esters) was higher in less porous matrices. LWAM absorbed the highest PCM mass (12.55 g), while NAM and RCAM had higher percentages of filled pores (up to 23.85 % and 28.92 %, respectively). PCM incorporation improved thermal inertia across all mortars, but there was a tradeoff between the rise in conductivity and thermal inertia. LWAM demonstrated the best overall performance, with the lowest thermal conductivity (1.04–1.15 W/m·K) and highest specific heat (921.92–1243.10 J/kg·K). PCM addition also increased the compressive and flexural strength of LWAM by up to 15 %, with less significant improvements observed in NAM and RCAM. This study's novelty is its fundamental understanding of the combined effects of PCM viscosity, mortar porosity, and PCM type on thermal and mechanical properties, contributing to the deployment of energy-efficient construction materials.

1. Introduction

Reducing greenhouse gas emissions is one of society's primary goals to minimize the effects of global warming. The International Energy Agency stated that 34 % of final energy consumption is due to the buildings sector, contributing to 37 % of total CO₂ emissions worldwide [1], and 40 % of the energy consumption of buildings comes from heating and cooling [2]. Under the current trend, the average surface temperature is expected to rise by 2.7 °C in 2100 [3]. In consequence, governments and organizations proposed strategies to reduce energy consumption like the Green Deal [4], the European Climate Law [5], and the 2030 Agenda for Sustainable Development [6].

Latent heat thermal energy storage reduces energy demand, improves the efficiency of HVAC systems, and tackles the mismatch between energy demand and supply [7]. This can be achieved with the

incorporation of phase change materials (PCMs) in buildings [8]. PCMs store and release latent thermal energy when they incur in a phase change process, which might be a modification of the state of aggregation (solid-liquid, liquid-gas, solid-gas) or an internal molecular structure rearrangement (solid-solid or liquid-liquid) [7]. Solid-liquid PCMs are preferred for building applications because their latent energy is higher than same-phase PCMs, while the volume variation between the phases is reduced [9]. Thus, PCMs store thermal energy when they are heated and melt, and they release the stored energy when they are cooled and solidify. While PCMs undergo the phase change process, their temperature is almost constant, creating isothermal energy reservoirs, increasing the thermal inertia of building envelopes, reducing energy consumption, and improving thermal comfort inside of the building [10].

PCMs are categorized according to their chemical composition into

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organic (paraffins and non-paraffins), inorganic (salt hydrates, metals, and alloys), and eutectic mixtures [11]. The desired properties of PCMs for building envelope applications are high latent heat and specific heat capacity, reduced volume changes, non-corrosiveness, non-toxicity, non-flammability, chemical stability, absence of subcooling, and a reasonable price and availability on the market [12]. Although no PCM fulfills all these requirements, organic PCMs are the fittest for construction applications [13].

The incorporation of PCMs into cementitious materials recently gained attention since they might be used to manufacture active PCM walls, floors, and roofs [14]. However, the phenomenological interactions between PCMs and cement are still not fully understood, and more research is needed [15].

The simplest method is directly incorporating the PCM while mixing cement products, but it is not an effective method due to leakage, chemical reactions, or microstructure modification [16]. Other techniques were proposed to overcome this, such as shape-stabilization and microencapsulation.

Shape-stabilization consists of containing the PCM in the pores of a supporting material, which is then added to cementitious materials. The supporting materials comprise clay-based powders like diatomite [17] or montmorillonite [18], carbon-based materials like expanded graphene [19] or biochar [20], and lightweight fine aggregates. Esmaeeli *et al.* studied the impregnation of lightweight aggregate with paraffin oil. They analyzed mortar and concrete with expanded shale soaked in liquid PCM for 72 hours. They studied the freezing-thawing behavior of the resulting mixtures and found that the thermally active cementitious materials reduced the freezing time by at least 10 % [21]. Aguayo *et al.* used expanded slate, expanded shale/clay, perlite, and pumice as PCM carriers for cementitious composites. They soaked the aggregates in Puretemp 24X (biobased PCM) for 72 hours. They found that 5 % PCM in the volume of the samples reduced the conductivity by more than 10 %. They also observed that pore diameter, in addition to porosity, conditions the PCM absorption [22]. Mohseni *et al.* developed structural concrete with PCM-incorporated lightweight aggregates. They vacuum-impregnated the lightweight aggregates with PureTemp23 (organic PCM) and coated them with epoxy resins and silica fume. Thermal performance tests indicated that the developed concrete could reduce energy consumption thanks to the thermal stabilization it provides [23]. Adding coating materials like epoxy resin enhances the compressive strength of cementitious materials since it prevents leakage of the PCM from the pores of lightweight aggregates. However, this technique is considered impractical because it is time-consuming and complex [24].

Microencapsulation prevents PCM leakage and provides a heat exchange surface, improving the thermal behavior. It is based on enclosing PCMs in thin, high-molecular-weight polymeric films in the shapes of spheres or rods [25]. Liu *et al.* demonstrated that fly ash mortar with microencapsulated PCM can be 3D printed. Two mixtures were studied, with 1 wt% and 3 wt% of PCM. Based on numerical simulations of the developed PCM mortar, they found that its thermal performance led to 30 % savings in building energy consumption [26]. Cabeza *et al.* analyzed the thermal behavior of a house-like cubicle of conventional concrete with microencapsulated PCMs a decade after its construction. Micronal DS 5001 PCM was employed, and the mixtures contained 5 wt % of PCM. The thermal enhancement thanks to the PCMs was not degraded with time, and thermal oscillations were reduced [27]. Shi *et al.* studied the thermal and mechanical properties of lightweight aggregate mortar with microencapsulated paraffin Micronal DS 5040X. They added up to 25 wt% of PCM microcapsules and found that higher PCM contents increased the heat capacity of the mortars. However, more PCM increased the average void size and the air content in the samples, thus reducing the thermal conductivity and effusivity [28].

PCM incorporation was analyzed in the cases of conventional and lightweight cementitious materials. Recycled concrete aggregates have also been assessed in cement composite mixtures, seeking to increase the

sustainability of the construction industry. Klemczak *et al.* studied the use of waste foam concrete, along with microencapsulated PCMs, in cement mortars [29]. Suttaphadkee *et al.* incorporated paraffin into recycled concrete aggregate through vacuum. Subsequently, they used the PCM-aggregate composite in lightweight concrete [30]. Rostami *et al.* proposed the use of paraffin-coated recycled concrete aggregates [31].

Several PCM incorporation methods of PCM into cementitious materials were explored in the available literature. These works reported an increase in cementitious mixtures' thermal energy storage capacity. However, they also identified drawbacks. While direct incorporation is the simplest and most cost-effective method, it has significant drawbacks: PCM leakage during phase transitions, potential material incompatibilities, and adverse effects on the mechanical properties and long-term stability of the composite [32]. It might also lead to worsening the workability of the mixtures. In the case of PCM-impregnated aggregates, compressive and flexural strength are reduced due to the reduction of adhesion between the PCM-treated aggregates and the cement matrix [33]. In the case of microencapsulated PCMs, the reduction of mechanical properties was also observed due to the modification of the hydration of the composites and the evolution of the pore microstructure [15]. Besides, PCM microcapsules might be damaged during the mixing process, breaking the films and leading to leakage of the core PCM. This hinders the improvement of the thermal performance of cementitious materials with microencapsulated PCMs [34]. In contrast, vacuum impregnation emerges as a promising alternative for addressing all these issues. This method creates a vacuum environment that facilitates PCM penetration into porous materials, preventing leakage during phase transitions [35]. As research continues, vacuum impregnation is increasingly recognized as a viable solution for overcoming the challenges of PCM integration. However, more research is needed to make it more accessible and cost-effective for a broader range of constructions [36].

To address these drawbacks, a novel PCM incorporation method, well suited for precast production elements, was proposed in a recent study [37]. In this method, the PCM is incorporated through vacuum into the pore structure of hardened cementitious composites. The results showed that this method enhances the composite's effusivity without compromising its compressive strength.

While the results of this recent study are very promising, this new work goes further by addressing two critical aspects that have received little attention: the role of PCM properties and the porosity of aggregates in the effectiveness of this novel incorporation method. Specifically, this study fills a gap in the literature by comparing the incorporation of different types of PCMs (one paraffin and two biobased esters) into three types of mortars, defined by their fine aggregate: natural, recycled concrete, and lightweight. The effect of the interaction between the mortar porosity and the PCM viscosity is analyzed, as it directly influences the amount of PCM that can be absorbed and the thermal performance of the mortar with PCM. The novelty of this work is analyzing how different PCM types impact the thermal and mechanical properties of mortars with various aggregates. This assessment is critical for defining a standardized procedure for PCM incorporation as a function of the composite's properties and the behavior of the PCM used, aiming at practical implementation in both precast and broader construction applications.

2. Materials and methods

2.1. Materials

The analyzed mortars were manufactured with CEM II cement (CEM II/A-V 42.5 R), water, fine natural silica sand 0 – 2 mm (sand-0/2) and fine aggregate 0 – 4.75 mm. This study assessed three types of 0 – 4.75 mm aggregate (natural silica aggregate NA-0/4.75, recycled concrete aggregate (RCA-0/4.75), and lightweight aggregate (LWA-0/

4.75), yielding three types of mortar: natural aggregate mortar (NAM), recycled concrete aggregate mortar (RCAM), and lightweight aggregate mortar (LWAM).

The natural aggregate in the NAM is silica sand. The recycled concrete aggregate in the RCAM was retrieved from a local quarry. The lightweight aggregate in the LWAM is expanded clay ARLITA® S [38]. It consists of granules with a porous core. The granulometry of these aggregates and the sand-0/2 is presented in Fig. 1. The lightweight aggregate is a commercial product whose granulometry is given by the manufacturer. However, the natural and recycled concrete aggregates come from local quarries. The NA and RCA granulometry was customized to resemble the lightweight aggregate as much as possible. The differences between the sieving curves of the different aggregates are due to the disparity between European and American sieves. These differences are negligible for the scope of this work.

Table 1 shows the mixing proportions of the studied mortars. In all cases, the volume proportion of the 0 – 4.75 mm aggregate was the same, and the water-to-cement ratio was 0.4. Therefore, the only difference between the mortars is the 0 – 4.75 mm aggregate type used. To design the mixtures of the different mortars, the dosage was done in volume of dry components per total volume of the mixtures. The effective water is the water added to the mixture without including the water sorption by the dry materials. The aggregates were oven-dried before the mixing process.

The mixing of the mortars followed the ASTM C305–20 standard [39]. For each mortar type, 18 samples were cast with $16 \times 4 \times 4$ cm dimensions. In total, 54 samples were manufactured. The samples were set in molds for 24 hours. The samples were covered with a plastic film to avoid water loss during the setting time in molds. After hardening, the samples were de-molded and placed into a wet chamber to cure for 28 days at 95 ± 5 % RH and 25 ± 1 °C. Once cured, the samples were dried in a 60 °C oven until the weight loss between two consecutive days was lower than 0.2 %. After drying, they were stored in laboratory conditions (50 ± 5 % RH and 21 ± 1 °C).

The PCMs that were incorporated into the mortars are RT21-HC (P21), CrodaTherm 21 (E21), and CrodaTherm 19 (E19). P21 is a paraffin mixture with a phase change temperature of around 21 °C manufactured by Rubitherm GmbH [40]. E21 and E19 are bio-based esters derived from plant-based feedstocks. They were produced by Croda International Plc. and have phase change temperatures of around 21 °C [41] and 19 °C [42], respectively.

2.2. Methods

The experimental campaign was designed to characterize the PCMs and the mortars with and without PCM. 18 samples of each mortar type were manufactured: 3 were used to determine the mortar density and

Table 1

Material quantities for 1 m³ of mixture.

Mixture	Cement (kg)	Fine sand (kg)	NA (kg)	RA (kg)	LWA (kg)	Effective Water (kg)
NAM	732.47	920.90	335.34	0.00	0.00	292.99
RCAM	732.47	920.90	0.00	280.61	0.00	292.99
LWAM	732.47	920.90	0.00	0.00	247.60	292.99

permeable porosity, 3 for the pore size analysis, 3 for the thermal and mechanical characterization as reference samples without PCM, and the remaining 9 samples were used for PCM incorporation and characterization. Fig. 2 shows the scheme of the experimental campaign with the tests performed for the PCM and mortar characterization.

2.2.1. PCM characterization

The viscosity and thermal properties of the studied PCMs were analyzed since they affect the samples' overall thermal behavior.

A low-temperature differential scanning calorimetry (LT-DSC) was used to analyze the transient thermal behavior of the PCMs when they were heated and cooled. It compares the thermal response of the PCM sample and a reference material (with known properties) when they are heated and cooled at a constant rate [43]. The equipment used in this study was a TA Q2000 LT-DSC (TA Instruments), with a temperature accuracy of ± 0.1 °C and a calorimetric precision of 0.05 %. The tests were performed in an N₂ atmosphere.

The PCM sample sizes were 17.7 mg for P21, 20.5 mg for E21, and 19.8 mg for E19. The temperature variation rate was 1 °C/min, complying with the ASTM D3418 – 21 [44], which indicates that it should be between 0.5 °C/min and 20 °C/min. The choice of a low temperature variation rate was based on previous studies on DSC applied to PCMs, which state that low temperature variation rates must be used to obtain correct DSC measurements [34]. The minimum and maximum temperatures for the test were –10 °C and 50 °C, respectively. This ensures that both the solidification and melting of the PCMs are analyzed. LT-DSC yields the effective heat capacity $C_{p,eff}(T)$, phase change temperature range, the enthalpy temperature relationship $h(T)$, and the latent heat L . These properties determine the phase change behavior and latent heat exchange.

The thermal properties of the liquid PCM were evaluated using the Trident Thermal Conductivity Analyzer (C-Therm Technologies Ltd.), with a 5 % accuracy for the conductivity measurement. The modified transient plane source (MTPS) method was used. It employs a single-sided interfacial heat reflectance sensor that applies a momentary constant heat source to the PCM sample. The emitted heat produces a temperature difference in the sensor-sample interface, which leads to a voltage variation in the sensor. This variation is used to directly calculate the effusivity e_{PCM} and thermal conductivity of the PCM k_{PCM} , complying with ASTM D7984–21 [45]. The heat capacity was calculated according to Eq. (1), where the density of the liquid PCM ρ_{PCM}^l is known: 770 kg/m³ for P21, and 850 kg/m³ for E21 and E19.

$$C_{p,PCM} = \frac{e_{PCM}^2}{k_{PCM} \cdot \rho_{PCM}^l} \quad (1)$$

The viscosity of the three PCMs was measured using a Modular Compact Rheometer MCR302 (Anton Paar Inc.). This rheometer uses two parallel plates actuated by an air-supported motor. It controls the sample temperature through a high-precision Peltier temperature device. The tests were performed by controlling the shear rate applied to the sample. For each PCM, the shear rate was increased in 1 s^{–1} steps from 1 to 100 s^{–1}. Shear stress, applied torque, and viscosity μ_{PCM} were measured. These tests were done with a temperature of 34 °C to represent the actual viscosity of the PCMs during the incorporation tests.

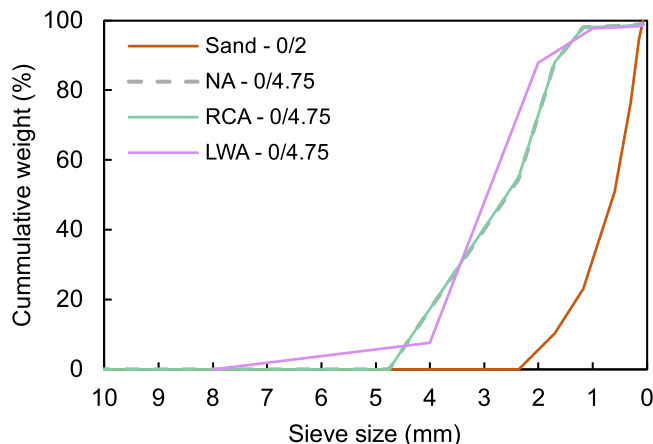


Fig. 1. Sieving curves of the aggregates.

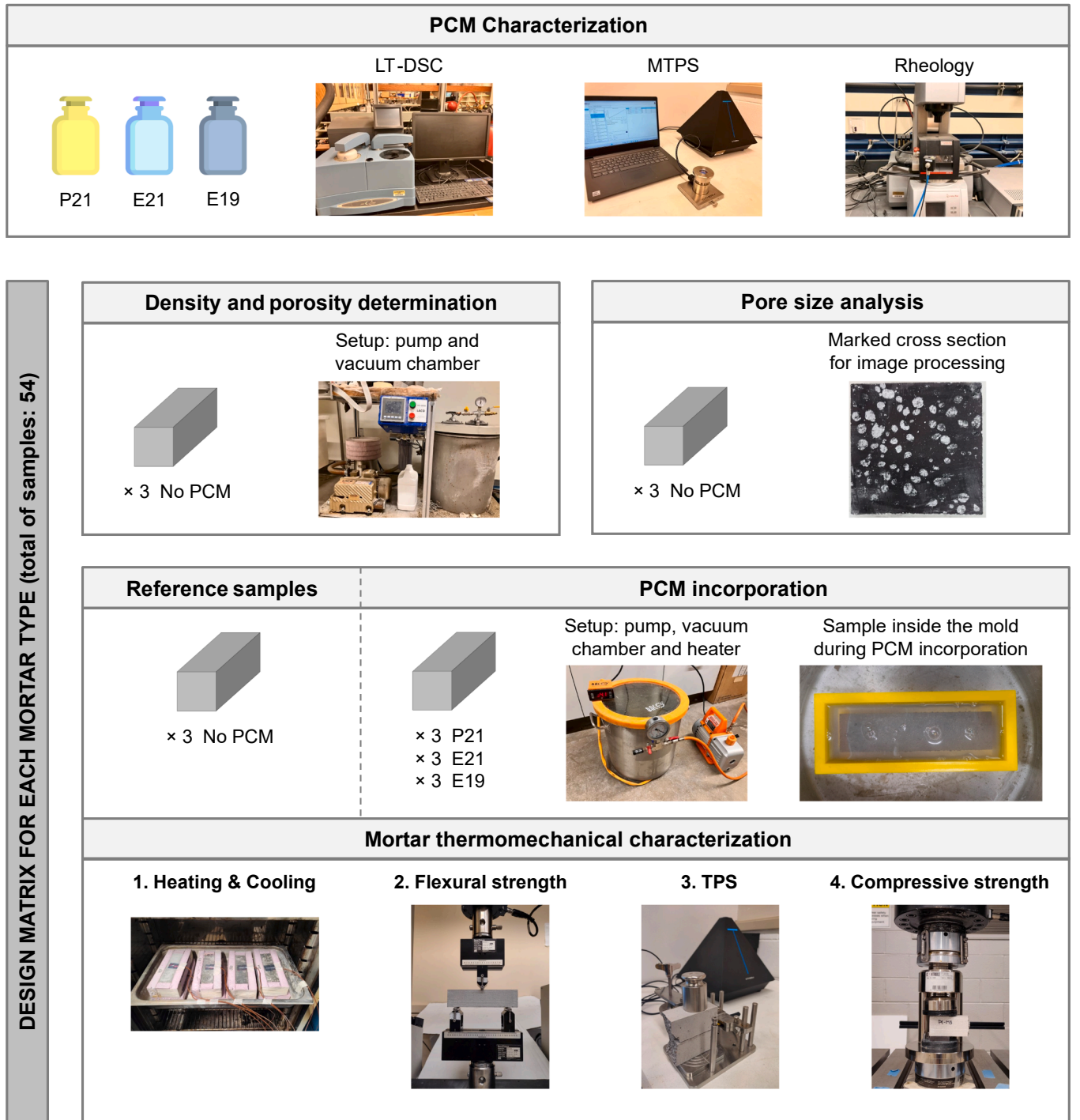


Fig. 2. Experimental campaign scheme.

2.2.2. Determination of the mortar density and permeable porosity

For each mortar type, 3 samples were tested. Before conducting the test, they were weighted, obtaining the dry weight m_D . Following the ASTM C830 – 00 [46], the oven-dry samples were placed inside a vacuum chamber, pumped down to an absolute pressure lower than 6.4 kPa and kept in that condition for 30 min. Then water was piped into the chamber until the samples were completely submerged, and vacuum was kept for 60 min. After saturating the samples in water, the suspended weight m_S and the saturated weight m_W were measured. The exterior volume of the samples V_{ext} was determined according to Eq. (2), and subsequently the initial porosity $P_0(\%)$, apparent density ρ_{App} , and bulk density ρ_B were calculated following Eqs. (3)–(5) respectively:

$$V_{ext} = (m_W - m_S) / \rho_w \quad (2)$$

$$P_0(\%) = \frac{m_W - m_D}{V_{ext}} \cdot 100 \quad (3)$$

$$\rho_{App} = \frac{m_D}{(m_D - m_S) / \rho_w} \quad (4)$$

$$\rho_B = m_D / V_{ext} \quad (5)$$

where ρ_w is the density of water. The apparent density is the density of the solid matrix (plus the impervious pores), while the bulk density is the

density of the composite considering the free porosity.

2.2.3. Pore size analysis

To analyze the pore size, 3 samples of each mortar were selected and cut in the plane of 4×4 cm cross-section with a Buehler IsoMet Low-Speed precision cutter. Subsequently, the cut surfaces were polished with Hillquist mesh flat diamond laps of 45, 30, and 15 μm . The polished surfaces were colored with a black felt-tip pen and the pores were filled with white barium sulfate powder. This process marks the pores in white color, while the remaining solid matrix is painted in black. The modified surfaces were photographed with a high-resolution camera (64 MP), and the images were processed in ImageJ software [47]. First, they were transformed from RGB into grayscale. Then, the threshold histogram limits were set as 100 and 255 to reduce noise. They were scaled, knowing the cross-section was 4×4 cm. Finally, the software automatically determined the pore size. This procedure has already been used in previous research [48].

2.2.4. PCM incorporation into the mortar samples

The PCM incorporation method is based on a previously published work by the authors [37], [49]. The setup for this process counts on three main elements: a vacuum chamber, a temperature-regulating heating plate, and a vacuum pump. The PCM incorporation method consists of the following steps:

- 1) Drying the samples until the difference in mass between two measurements performed 24 h apart is lower than 0.2 %.
- 2) Weighting the dried samples before the PCM incorporation.
- 3) Submerging the samples into a recipient that contains the PCM in the liquid state. The dry samples and the PCM are kept at 35 °C before starting the test, ensuring the PCM is liquid from the beginning of the test.
- 4) Placing the recipient with the PCM and the mortar sample inside the vacuum chamber. This chamber counts on a heating plate on its lower surface. Since the sample and the PCM must be at a temperature higher than the phase change temperature of the PCM to prevent it from solidifying, the temperature of the heating plate was kept constant at 35 °C.
- 5) Turning on the vacuum pump to begin the PCM incorporation. The pressure inside the chamber drops by 1 bar (750 mmHg). Once this pressure is stabilized, the vacuum is kept for 4 h. During this process, the air inside the pores of the mortar samples comes out from the solid matrix. Subsequently, the liquid PCM fills the emptied pores. The PCM impregnation begins at the samples' surface and propagates towards the center.
- 6) After 4 h of vacuuming, the pump is turned off, the chamber valves are opened, and the samples are removed from the recipient.
- 7) Finally, the PCM-impregnated samples are surface-dried and weighed.

The mass of PCM that was incorporated into each sample, m_{PCM} , is the difference between the mass of the sample before and after the incorporation of the PCM.

This method is restricted to a precast production setting [49]. The controlled environment of precast production and using automated systems facilitate consistent vacuum impregnation of PCM into the precast elements. This adaptability underscores the practical feasibility of vacuum impregnation for large-scale applications, as highlighted by recent advancements in precast manufacturing [50].

The PCM incorporation through vacuum was previously tested with cementitious composites [37], [49], and no leaking was observed thanks to the capillary forces that hold the PCM into the pores once vacuum is no longer maintained. This means that to extract the PCM from the pores an external force should be applied.

2.2.5. Mortar mechanical characterization

The mortar samples were tested to determine their flexural strength following the ASTM standard C348–21 [51]. The tests were performed using an MTS machine with a maximum load capacity of 10 kN, and a displacement control rate of 0.2 mm/min was used. The $16 \times 4 \times 4$ cm prisms were broken in halves due to this test. These halves were used first for the TPS tests and, subsequently, for the compressive strength tests.

The compressive strength of the mortar samples was determined following the ASTM standard C349–18 [52]. An MTS machine with a 300 kN maximum load capacity and displacement control. Its accuracy is 0.01 mm for the position and ± 0.05 % for the speed. Tests were done using a 0.05 mm/s displacement rate until the samples failed. 72 samples were tested using the $8 \times 4 \times 4$ cm half samples obtained in the flexural strength tests.

2.2.6. Mortar thermal characterization

The thermal properties of the mortar samples (with and without PCM) were determined using the transient plane source (TPS) method. This method uses the Trident Thermal Conductivity Analyzer. A double-sided hot disc sensor is placed between two identical samples. Power is applied to the sensor, and it heats up, producing a temperature difference in the interface with the samples. This variation is used to measure the thermal conductivity k , volumetric heat capacity (ρC_p), and diffusivity α of the samples, complying with the ISO standard 22007–2 [53]. The thermal effusivity of the composite e is calculated according to Eq. (6).

$$e = \frac{k}{\sqrt{\alpha}} \quad (6)$$

The specific heat capacity of the mortar samples without PCM is determined with Eq. (7), and with PCM, Eq. (8) is used. Since the TPS determines the volumetric heat capacity of the whole composite, it accounts for the solid matrix, air pores, and PCM. In consequence, the bulk density of the samples must be used to calculate the specific heat.

$$C_p = \frac{(\rho C_p)}{\rho_B} \quad (7)$$

$$C_p = \frac{(\rho C_p)}{\rho_B + m_{PCM}/V_{ext}} \quad (8)$$

Due to the heterogeneous nature of mortars, small samples required for DSC analysis may not accurately represent the overall thermal response of the material [54]. The complexity of the microstructure in concrete and mortars, characterized by a diverse range of components such as aggregates, cement paste, and air voids, poses challenges that limit the applicability of DSC for thermal analysis. That is why the TPS method was employed alongside heating-cooling tests. They analyze the thermal behavior of the entire mortar sample comprehensively, while they overcome the limitation of DSC for heterogeneous materials.

Additionally, non-standard heating and cooling tests were designed to assess the thermal inertia of the different mortars with and without the different PCMs. For the three types of mortar, 4 samples were studied: the reference sample without PCM and one sample with each PCM. In total, 12 samples were analyzed.

The samples were thermally insulated on each surface except for the top one. To do so, 2.54-cm-thick (1 in) XPS rigid foam was used. These samples were placed in an oven with a constant temperature of 35 °C until they reached a uniform and stable temperature. Then, they were removed from the oven and placed inside a cold chamber with a constant temperature of 10 °C for 3 hours. Finally, they were introduced back into the oven at 35 °C for 3 hours. This cooling-heating test was repeated 3 times for all samples. Two K thermocouples were placed on each sample: one on the top non-insulated surface and the other on the bottom insulated surface. The thermocouples were connected to a USB-TC 8-channel thermocouple input acquisition device (Measurement

Systems Ltd.). TraceDAQ software (Measurement Systems Ltd.) was used to visualize the temperature readings and to export the data. Fig. 3 shows the setup with the insulated samples, the upper thermocouples, and a scheme of the cross-section showing the location of thermocouples.

These tests measure the transient temperature in the top and bottom surfaces of the samples. The difference in these temperatures illustrates the thermal inertia. Mortars with higher inertia will exhibit more significant differences between top and bottom surface temperatures. These tests also identify the phase change of the incorporated PCMs. Since the thermocouples are placed on the surface of the samples, where the PCM was incorporated, they show the thermal stabilization while the PCMs undergo the phase change process.

3. Results and discussion

3.1. Thermal characterization of the PCMs

The LT-DSC results of the studied PCMs are shown in Fig. 4. The heat flow of the PCM samples measures their transient thermal response during melting and solidification. Bell-shaped curves characterize the phase change process, while the sections with constant heat flow indicate that the PCM is fully solid or liquid. The temperatures where the bell-shaped curves begin and end denote the phase change temperature range. P21 presents the lowest value of heat flow but also the broadest phase change temperature range (around 11.5 °C both in melting and solidification). The esters E21 and E19 exhibit narrower phase change ranges (up to 7.57 °C and 6.93 °C respectively).

These results also indicate the hysteresis effect in all PCMs. This is the phenomenon where the melting and solidification occur at different temperatures and with unequal thermal response [55].

The ester E21 shows subcooling. This means that its solidification requires an extra drop of temperature to begin (known as the degree of subcooling) [56]. When this happens, after solidification starts, the temperature rises suddenly, which is visible in Fig. 4.

Table 2 shows the phase change properties calculated from the LT-DSC results: latent heat, peak temperature, and phase change range. The latent heat of all PCMs is similar, around 200 kJ/kg. The peak temperature is the temperature where the heat flow is maximum. This is a relevant parameter when selecting a specific PCM for a given application. Finally, the phase change range of each PCM is indicated. The effects of hysteresis are observed in the discrepancies between melting and solidification in the peak temperature and the phase change range. The differences are more significant for the esters, complying with previous studies that indicate that hysteresis is more relevant in organic bio-based PCMs [57].

Table 3 presents the thermal properties of the PCMs obtained from the MTPS tests. Very close values of thermal conductivity were obtained in all PCMs. The specific heat capacity of the paraffin is higher than that of the esters. However, the paraffin presents the lowest effusivity since its density is also the lowest.

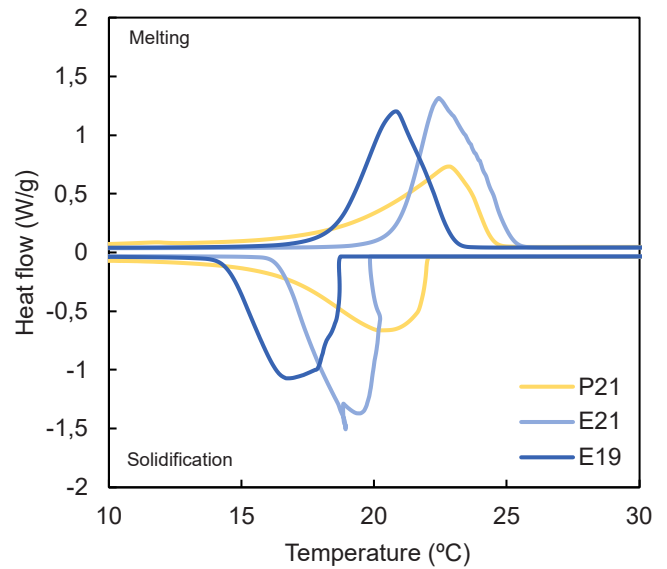


Fig. 4. DSC results of the PCMs.

3.2. PCM viscosity

The results of the viscosity tests at 34 °C are shown in Fig. 5. The resulting curves are characterized by a sharp viscosity increase for low shear rates, and then the viscosity stabilizes. Thus, the viscosity of the PCMs at 34 °C is set as the recorded value for a shear rate of 100 s⁻¹. The viscosity of the studied PCMs is 12.6 mPa·s for P21, 5.2 mPa·s for E21, and 6 mPa·s for E19. Results show that the paraffin's viscosity is twice the esters' viscosity.

3.3. Mortar physical properties

Pore size analysis tests quantify pores and determine pore size distribution in each mortar type, see Fig. 6. Since three samples of each type were tested, the average number of pores is represented by solid bars and the standard deviation by error bars. LWAM presents the highest number of pores regardless of the pore size range. The pore size distribution is similar in NAM and RCAM. This complies with the results of the initial porosity, where LWAM exceeded NAM and RCAM by far. However, NAM displays more pores smaller than 0.005 mm², while RCAM shows more pores bigger than 0.005 mm². Mortars with NA have the most pores in the cement matrix, reducing their potential size. On the contrary, the porosity of a mortar with RCA is significantly influenced by the presence of adhered cement paste from the original concrete, typically more porous and less dense than the matrix of natural aggregates [58]. The presence of microcracks also influences this due to the weakened interfacial transition zone in mortars with RCA [59].

The average pore size of each mortar, A_p , was determined as the average of the pore sizes of the three analyzed samples, shown in



Fig. 3. Transient thermal characterization setup: (a) general disposition, (b) position of the thermocouples.

Table 2

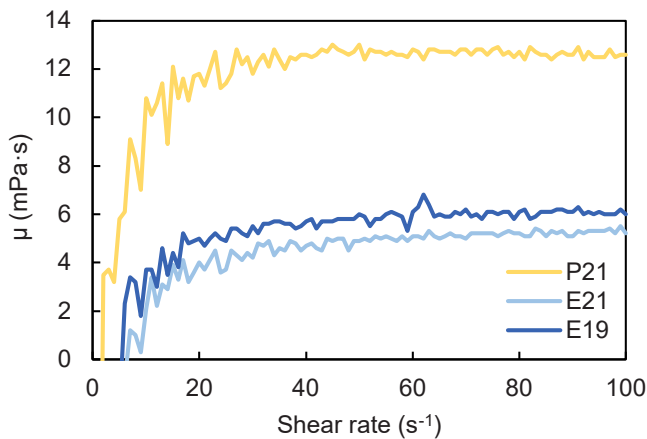
Phase change properties derived from the LT-DSC tests.

	Latent heat (kJ/kg)			Peak temperature (°C)			Phase change range (°C)		
	P21	E21	E19	P21	E21	E19	P21	E21	E19
Melting	200.73	201.65	205.79	22.84	22.44	20.82	13.58 - 24.97	18.14 - 25.71	16.51 - 23.44
Solidification	206.48	191.29	209.52	20.21	18.92	16.73	10.48 - 22.12	15.66 - 20.1	13.44 - 18.88

Table 3

Thermal properties of the PCMs.

	P21	E21	E19
k_{PCM} (W/m·K)	0.158	0.161	0.161
Cp_{PCM} (J/kg·K)	2131.2	1976.3	1978.2
e_{PCM} (W·s ^{1/2} /K·m ²)	508.8	520.0	520.5

**Fig. 5.** Results of the viscosity tests of the PCMs at 34 °C.**Table 4.** The initial porosity and density of the mortars are also presented.

LWAM samples present the highest initial porosity (33.12 %), while NAM and RCAM samples show lower and similar values of porosity (17.68 % and 19.01 %, respectively). This is due to the type of aggregate used for each mixture, where the lightweight aggregate is characterized by its high porosity, followed by the recycled concrete aggregate.

Similarly, the average pore size of LWAM is the highest of all. The apparent and bulk densities have the opposite tendency: the highest values are found in the NAM samples.

3.4. PCM incorporation

The results of the PCM incorporation are shown in Fig. 7. The pattern fill indicates the percentage of the volume of the mortar samples filled with PCM, $V_{PCM}(\%)$, while the solid fill is the remaining porosity after the PCM incorporation. $V_{PCM}(\%)$ was calculated according to Eq. (9) and the average values in each case are included in Fig. 7.

$$V_{PCM}(\%) = \frac{m_{PCM}/\rho_{PCM}^I}{V_{ext}} \cdot 100 \quad (9)$$

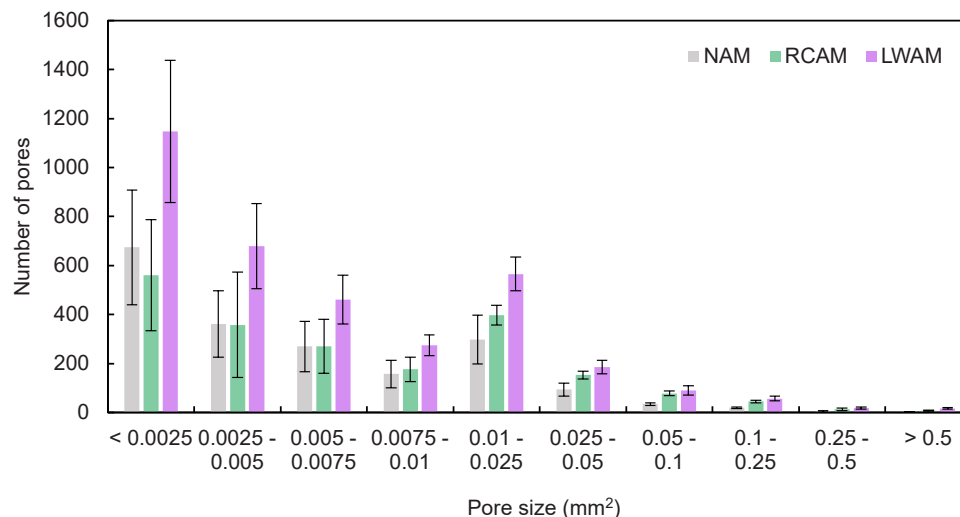
NAM presents the lowest volumes of incorporated PCM, ranging from 1.37 % with P21 to 4.22 % with E21. In RCAM, the PCM incorporation was higher, comprising 2.42 % with P21 and 5.54 % with E19. LWAM shows the highest volume of incorporated PCM, ranging from 4.33 % with E19 to 6.37 % with P21. This indicates that the higher the initial porosity of the mortar, the greater the volume of PCM incorporated since a higher initial porosity implies more pores that might be filled with PCM.

However, the PCM that leads to the highest absorbed volume is not the same in all mortars. The incorporated volume of esters is similar for each mortar type, but the paraffin follows a different trend. In NAM and RCAM, P21 yields the lowest incorporation. In LWAM, the maximum

Table 4

Physical properties of the different studied mortars.

Mortar type	A_p (mm ²)	P_o (%)	ρ_{App} (kg/m ³)	ρ_B (kg/m ³)
NAM	0.0136	17.68	2545.30	2095.18
RCAM	0.0272	19.01	2520.40	2040.84
LWAM	0.0584	33.12	2410.84	1612.36

**Fig. 6.** Pore size distribution of the different studied mortars.

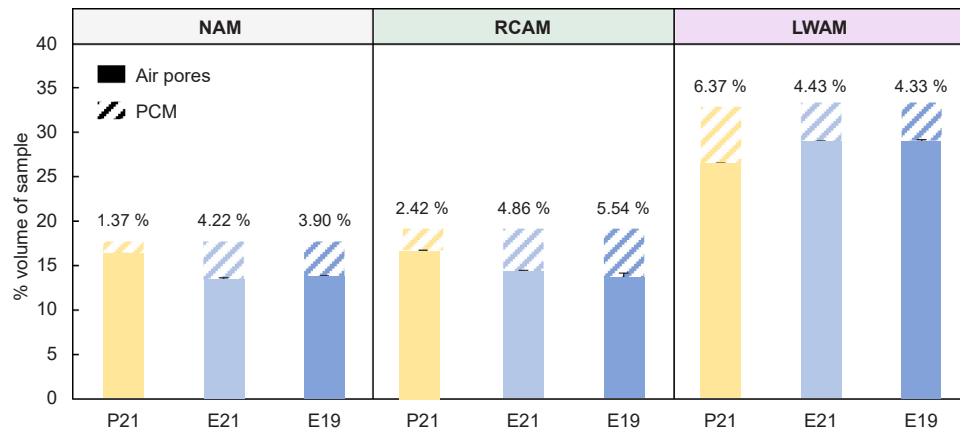


Fig. 7. Percentage of the volume of the mortar samples filled with PCM (specified for each case) and remaining pores filled with air.

PCM incorporation happens for P21. This unequal tendency indicates a correlation between the initial porosity, the pore size distribution, the type of PCM, and the absorbed quantity of PCM.

The mass of incorporated PCM and the percentage of volume of the samples filled with PCM are shown in Table 5. The incorporated mass of PCM ranged from 2.693 g to 12.553 g and it follows the same trend as the percentage volume: on average, the lowest values are found in NAM, and the highest values in LWAM. Table 5 also shows the percentage of the initial porosity that has been filled with PCM, V_{PCM}/P_0 (%). Even if LWAM presents the highest absorbed volume of PCM, the ratio V_{PCM}/P_0 is greater in NAM and RCAM, on average. Thus, up to 23.85 % of the pores of NAM are filled with PCM, and up to 28.92 % for RCAM, while in LWAM the highest value of this ratio is 19.37 %. This is explained by the higher value of P_0 in LWAM. These results indicate a correlation between the initial porosity, the pore size distribution, the type of PCM, and the absorbed quantity of PCM.

Fig. 8 presents the cross-sections of the different mortar samples with PCM. These cross-sections were obtained after performing the flexural strength tests. For better visualization, the original photos of the cross-sections were edited to brightness -19, contrast +26, lights -100, shadows -16, and saturation +27.

The cross-sections of the reference mortar samples show the differences in the aggregates. NAM and RCAM exhibit a heterogeneous aggregate composition because natural sand and recycled concrete were retrieved from a local quarry. In contrast, the lightweight aggregate is more homogeneous because it is all expanded clay.

The cross-sections of the samples with PCM clearly show how the PCM is incorporated only in the external volume of the samples. This is more evident in the samples with E21 and E19 for all mortars and LWAM with P21, where the PCM-impregnated area is darker. The intensity of the PCM contour is proportional to the absorbed mass of PCM, but it is also related to the type of PCM itself. Thus, the esters yield darker contours, while in the paraffin, the contour is only detectable with LWAM, where the incorporated mass of PCM is enough to be seen.

Table 5

Average incorporated mass of PCM and average volume of the samples filled with PCM in percentage.

Mortar	PCM	m_{PCM} (g)	V_{PCM} (%)	V_{PCM}/P_0 (%)
NAM	P21	2.693	1.37	7.73
	E21	9.177	4.22	23.85
	E19	8.487	3.90	22.06
RCAM	P21	4.773	2.42	12.64
	E21	10.573	4.86	25.36
	E19	12.060	5.54	28.92
LWAM	P21	12.553	6.37	19.37
	E21	9.640	4.43	13.28
	E19	9.420	4.33	12.98

To provide a better understanding of the correlation between the incorporated mass of PCM m_{PCM} in the studied mortars, the initial porosity of the mortars P_0 (%) and the viscosity of the PCMs μ_{PCM} , a multiple linear regression model was developed. This is a statistical technique that analyzes how a variable of interest is affected by other independent variables [60]. In this case, the interest variable is m_{PCM} , and the influence of μ_{PCM} and P_0 was studied. From the experimental data, the regression model was built and the least squares method was used to estimate its parameters. In this case, a first-order model that takes into account the interaction between the independent variables was used, see Eq. (10), where $\hat{m}_{PCM}$ is the predicted incorporated mass of PCM, b is the intercept, a_μ is the coefficient of the viscosity, a_{P_0} is the coefficient of the porosity, and $a_{\mu P_0}$ is the coefficient of the interaction. A total of 27 [μ_{PCM}, P_0, m_{PCM}] points were used to build the model (3 samples of each type of mortar-PCM combination).

$$\hat{m}_{PCM} = b + a_\mu \mu + a_{P_0} P_0 + a_{\mu P_0} \mu P_0 \quad (10)$$

The results of the model fitting are presented in Table 6. The coefficient of determination of the model R^2 is 0.8415, and the adjusted R^2 is 0.8201, indicating the adjusted model effectively explains the relationship between the variables. The regression model also provides the statistical significance of each predictive variable through the P-Value. A P-Value below the 0.05 significance level (P-Value < 0.05) implies that the independent variable is statistically significant in explaining the variation in the predicted variable. The P-Values of a_μ , a_{P_0} and $a_{\mu P_0}$ are all lower than 0.05, revealing that μ_{PCM} and P_0 significantly influence m_{PCM} independently and jointly.

Fig. 9 shows the regression model results (gray surface), and the actual points used to create the model. The porosity has a limited effect on the esters, whose viscosity ranges from 5.2 mPa·s to 6 mPa·s. This is observed in the almost constant slope of the response plane for low viscosity values. In the case of the paraffin, whose viscosity is 12.6 mPa·s, the porosity is more relevant: the higher the porosity, the greater the PCM incorporated mass.

Thus, in low-porosity mortars (NAM and RCAM), using high-viscosity PCMs is detrimental to the incorporation. In contrast, in high-porosity mortars (LWAM), although PCM viscosity has less influence, the PCM incorporation is maximized with higher viscosity PCM. This means that porous mortars are suitable for PCMs with a large range of viscosity. Meanwhile, for low-porosity mortars, using PCMs with low viscosity is critical to optimizing the absorption of PCM. This highlights that PCM incorporation is not solely determined by viscosity or porosity alone but also by their interaction. For instance, while lightweight aggregate mortars exhibit higher porosity, the absorbed PCM volume varies depending on the PCM type, underscoring the importance of this analysis for practical applications.

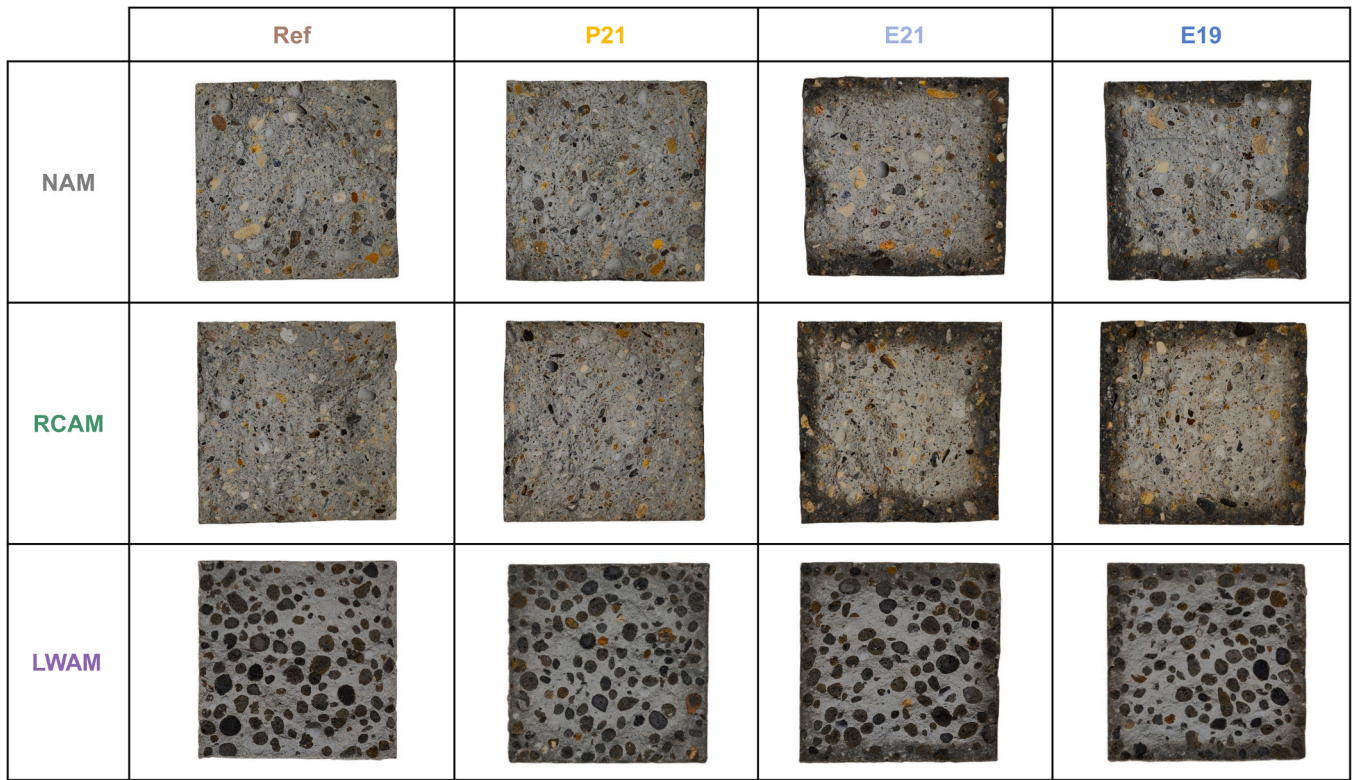


Fig. 8. Cross-section of the different mortars with and without PCM and average absorbed mass of PCM.

Table 6
Multiple linear regression model results.

Coefficient	Estimate	P-Value
b	24.656	3.58×10^{-10}
a_{μ}	-2.5506	3.22×10^{-9}
a_{P_0}	-0.52671	1.80×10^{-5}
$a_{\mu P_0}$	0.090249	4.93×10^{-8}

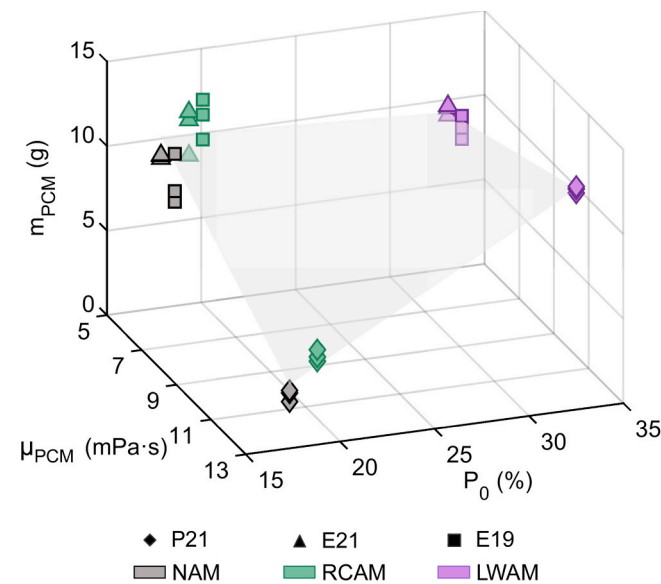


Fig. 9. Predicted incorporated mass of PCM obtained by multiple linear regression and measured points.

3.5. Mechanical properties

The average flexural strength and compressive strength of the studied mortars are presented in Fig. 10. In the cases without PCM, flexural and mechanical strength are inversely proportional to the porosity of the aggregate. Thus, the highest values are found for NAM and the lowest values for LWAM. More specifically, the flexural strength of RCAM and LWAM is 13.32 % and 38.12 % lower than in NAM, respectively. The compressive strength of RCAM was 13.25 % lower than in NAM, and in LWAM it was 51.40 % lower.

For the reference cases, a band with the amplitude of the variation was added. This helps noticing if the effect of the PCM incorporation in the mechanical properties of mortars with PCM is relevant. Values of flexural or compressive strength outside the bands of the reference mortars indicate that the effect of adding PCM is relevant. The values of the average difference in flexural strength and compressive strength are included in Fig. 10 too. PCM incorporation does not significantly affect the flexural strength of NAM and RCAM, but in LWAM, it leads to a substantial increase. This increment happens because, in LWAM, the incorporated mass of PCM was the greatest, and this type of mortar initially had the lowest flexural strength. Similar trends are observed in compressive strength results. This reveals that PCM incorporation does not compromise the mechanical properties of the studied mortars.

It is also highlighted that after the compression tests no PCM leaked the samples, demonstrating that the vacuum-incorporation method prevents this issue.

3.6. Thermal properties

The incorporation of PCM into the pores of the mortar seeks to increase thermal inertia. However, PCMs have a thermal conductivity up to 10 times higher than the thermal conductivity of the air they are substituting. As such, the incorporation of PCM presents a tradeoff between the increase of the thermal inertia (effusivity and specific heat capacity) and the increase of the conductivity of the composite mortar-

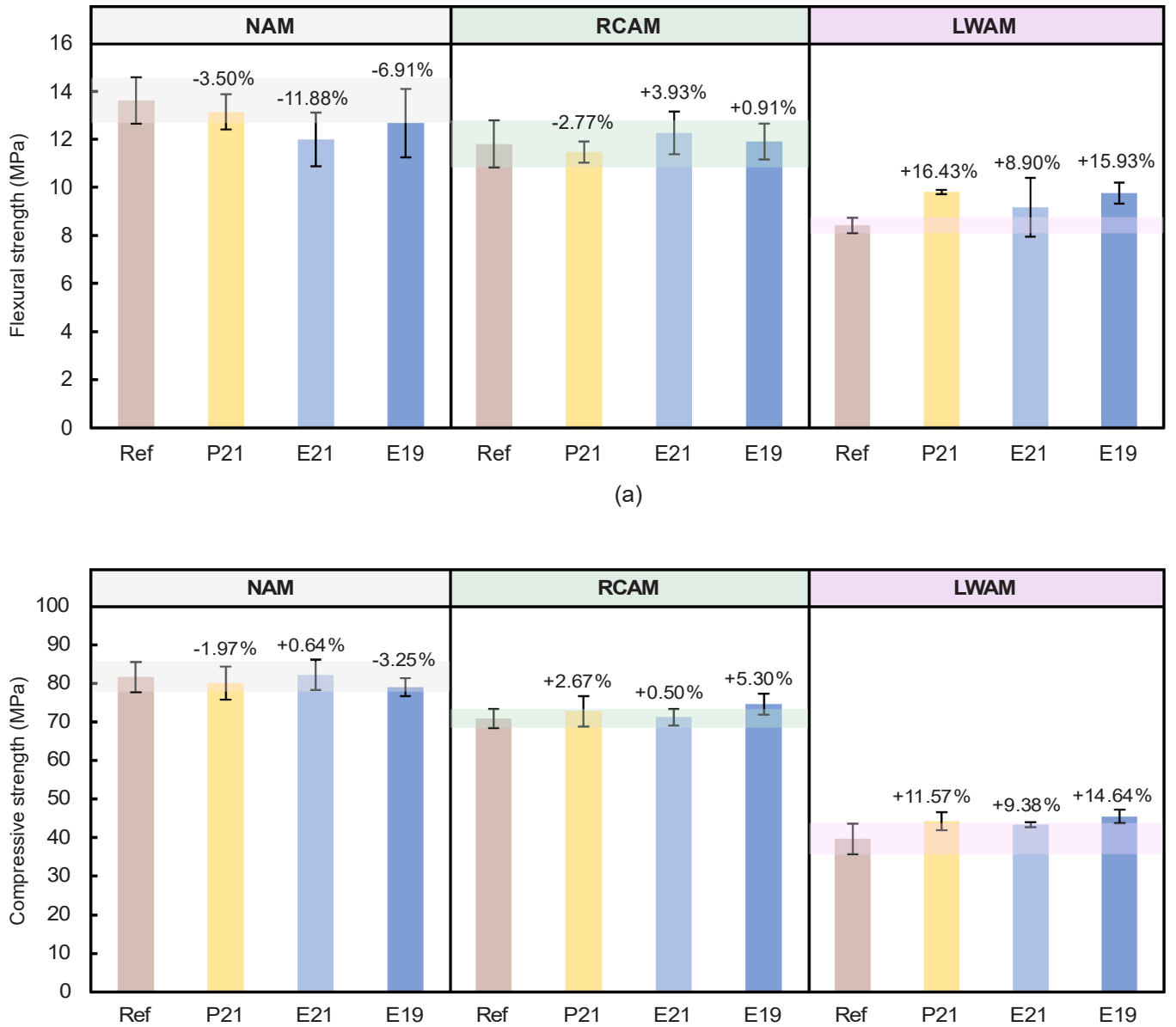


Fig. 10. Mechanical properties of the studied mortars: (a) flexural strength and (b) compressive strength.

PCM.

Fig. 11 presents the mortar samples' conductivity, effusivity, and specific heat capacity. The results show a correlation between the conductivity and the other two properties. Eq. (11) shows the expected specific heat capacity depending on the conductivity of the reference sample $\hat{C}_{p_{ref}}$, and Eq. (12) shows the expected effusivity $\hat{e}_{ref}$. These equations were obtained by fitting the data of the samples without PCM to a second-order polynomial equation using the least squares method. They are represented in Fig. 11 by the dotted lines.

$$\hat{C}_{p_{ref}} = -112.82 \cdot k^2 + 645.51 \cdot k - 178.11 \quad (11)$$

$$\hat{e}_{ref} = -1169.45 \cdot k^2 + 4189.36 \cdot k - 1981.09 \quad (12)$$

The gray area, built with curves parallel to Eqs. (11) and (12), includes the results of all the reference samples. Both the specific heat capacity and the effusivity increase when the conductivity rises. For the reference samples, LWAM shows the lowest conductivity, effusivity, and specific heat. This is explained by the high porosity of this type of mortar and the insulating effect of the air. In contrast, NAM presents the highest

conductivity, effusivity, and specific heat. This is due to the lower porosity of this type of mortar. RCAM showed in-between values of the thermal properties since its porosity is lower than in NAM but higher than in LWAM. However, RCAM presented more dispersion in the results due to the higher heterogeneity of the aggregate.

Focusing on the results of the mortars with PCMs, it is observed that they show higher conductivity, effusivity, and specific heat than the corresponding reference mortar. As shown in Fig. 11, in LWAM and NAM, the increase of specific heat capacity is higher than the increase of conductivity with all the PCMs. In RCAM, this trend is not as clear due to the higher variability of the samples. Besides, in RCAM with P21, the specific heat did not increase with the increase in conductivity. Therefore, in NAM and LWAM with PCM, the increase in thermal inertia is greater than the increase in conductivity. This situation is the desired case because it implies that, even if the PCM increases the conductivity, the augmented thermal inertia compensates for it. More specifically, LWAM is the best solution because it reaches the highest values of specific heat with minimum conductivity.

The effusivity shows a similar trend but with less significant increments. This happens because of the definition of effusivity, as it

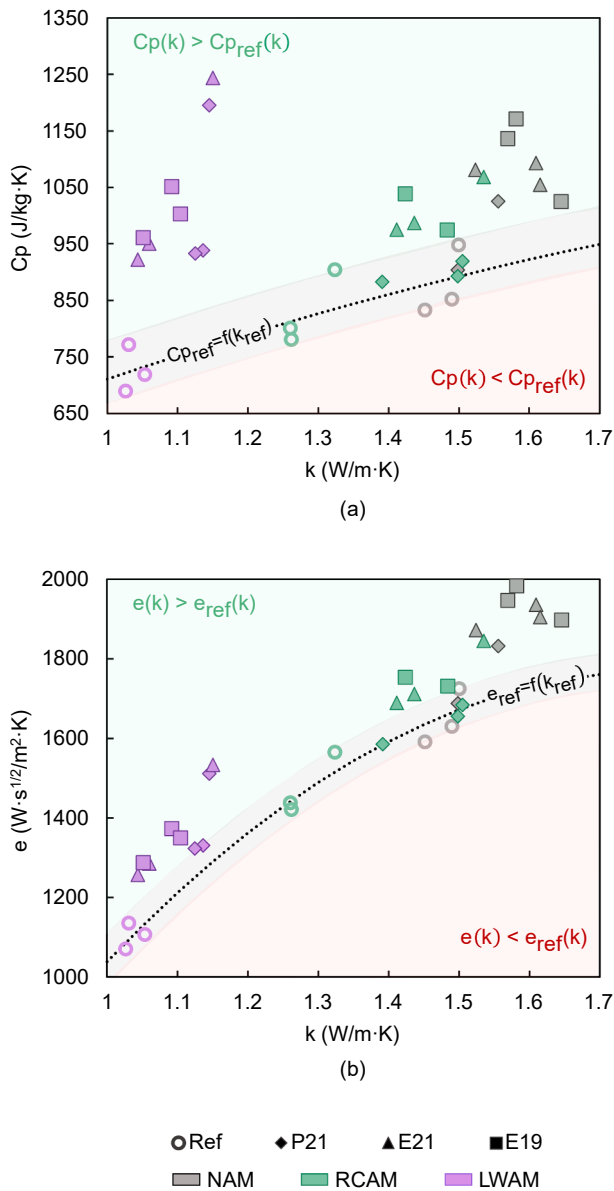


Fig. 11. Effect of PCM incorporation on thermal properties: (a) specific heat capacity, (b) effusivity.

depends on the square root of the product of several variables (conductivity, density, and specific heat).

To better understand the tradeoff between thermal inertia and conductivity, the percentage increase of conductivity, specific heat capacity, and effusivity was calculated and shown in Table 7. The average values for each mortar-PCM couple were computed with respect to the

Table 7

Percentage increase of conductivity, effusivity, and specific heat of the mortars with PCM.

Mortar	PCM	Δk (%)	ΔC_p (%)	Δe (%)
NAM	P21	3.17	9.86	6.71
	E21	6.93	22.61	15.49
	E19	8.02	26.49	17.76
RCAM	P21	14.28	8.43	11.32
	E21	13.98	21.92	18.55
	E19	13.44	21.45	18.11
LWAM	P21	9.51	40.63	25.74
	E21	4.58	42.92	23.03
	E19	4.44	38.25	21.02

reference case. The cases where $\Delta C_p(\%)$ and $\Delta e(\%)$ are greater than $\Delta k(\%)$ are desirable and correspond to the green region in Fig. 11. NAM showed the lowest increase of conductivity (3.17 – 8.02 %), followed closely by LWAM (4.44 – 9.51 %), while this increase was more significant in RCAM (13.44 – 14.28 %). Regarding the specific heat capacity, LWAM showed the highest improvement (38.25 – 42.92 %), while NAM and RCAM exhibited a more modest rise (9.86 – 26.49 % and 8.43 – 21.92 %, respectively). Only in RCAM with P21 did the conductivity increase outscore the thermal inertia increase. As a result, this combination is not suitable for building envelope applications.

3.7. Transient thermal response of the mortar samples

The transient thermal response of the studied mortars is characterized by the evolution of the temperature measured on the top and bottom surfaces. The transient temperatures of the top and bottom surfaces of the studied samples are presented in the following figures: Fig. 12 for NAM, Fig. 13 for RCAM and Fig. 14 for LWAM.

The results of the top and bottom surface are similar, characterized by the asymptotic tendency towards 10 °C in the cooling stage and towards 35 °C in the heating stage. The temperature difference between the top surface and the air drives the convection heat transfer. As the top surface heats up or cools down, conduction heat transfer occurs in the mortar. This implies that the bottom surface is insulated from the top by the mortar itself, explaining why the temperature variation is much lower at the bottom than at the top. Higher thermal inertia results in slower temperature variations at the bottom surface, as observed in mortars with PCM, especially for LWAM with higher PCM content.

The samples without PCM show the same behavior: an asymptotic response tending to 10 °C in the cooling test and to 35 °C in the heating test. The slope of the curves depends on the specific heat capacity, conductivity, and density. The higher the conductivity, the higher the slope of the thermal response and the faster it reaches 10 °C or 35 °C. Density and specific heat have the opposite effect: higher values imply a lower temperature variation.

In samples with PCM, the transient thermal measurements reveal the influence of the phase change process. Latent heat is exchanged during phase transitions, significantly increasing the thermal inertia. This effect stabilizes the temperature and slows down its variation. The duration and impact of this stabilization depend on the PCM type and the incorporated mass. The phase change in NAM and RCAM with P21 is barely noticeable due to the low PCM content. In contrast, the esters show more explicit phase change processes, with a clear stabilization during solidification and melting. For E19, the phase change occurs faster due to the higher initial temperature difference, leading to a more abrupt release of latent heat. Among all tested mortars, LWAM with P21 demonstrates the most evident effect of PCM incorporation due to its higher PCM mass. This highlights how the combination of mortar type and PCM affects the thermal response. LWAM samples with esters exhibit behaviors comparable to NAM and RCAM, although the phase change effects are more pronounced in LWAM due to the higher absorption of PCM.

Regarding the temperature difference between the top and bottom surfaces, these results underscore the significant role of thermal inertia in reducing temperature fluctuations. While the effect of P21 is minimal in NAM and RCAM, its influence is more pronounced in LWAM due to the higher mass of PCM. These findings suggest that the effectiveness of the systems depends on the target thermal performance and the specific conditions of use. LWAM with P21 or esters exhibits the highest potential for applications requiring enhanced thermal inertia, while systems with lower PCM content (e.g., NAM with P21) may offer limited thermal benefits under similar conditions.

These results show how incorporating PCM into the mortars creates thermally active materials. The samples with PCM show a dynamic thermal response that depends on the material's temperature and on the absorbed mass of PCM (which depends on the mortar type and PCM).

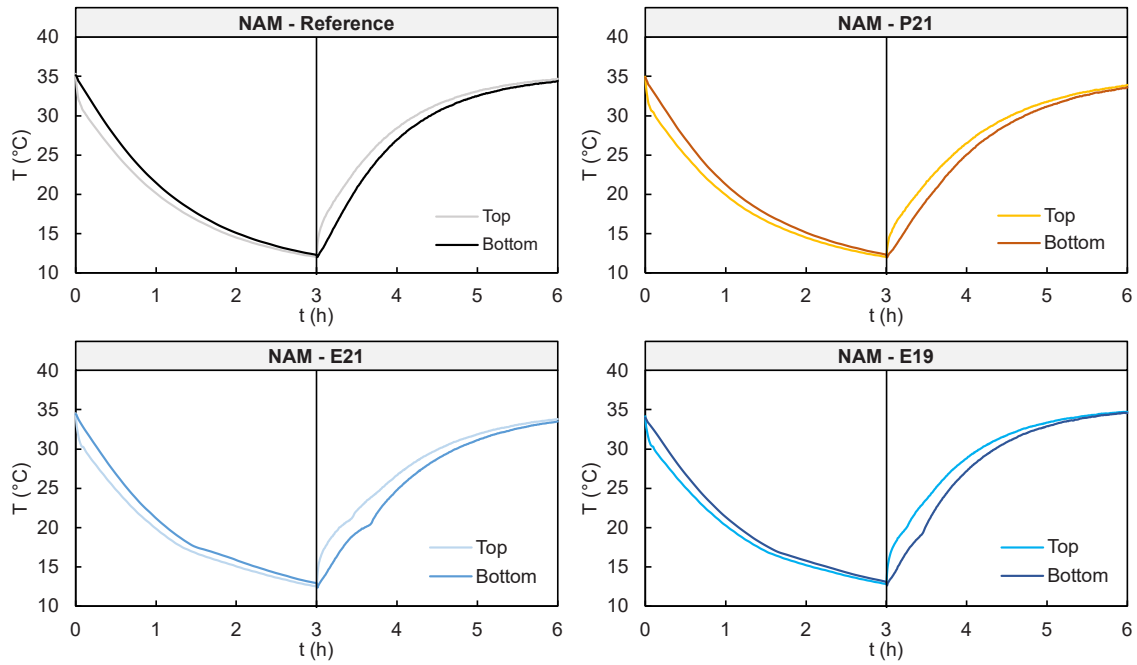


Fig. 12. Transient top and bottom temperature in the NAM samples.

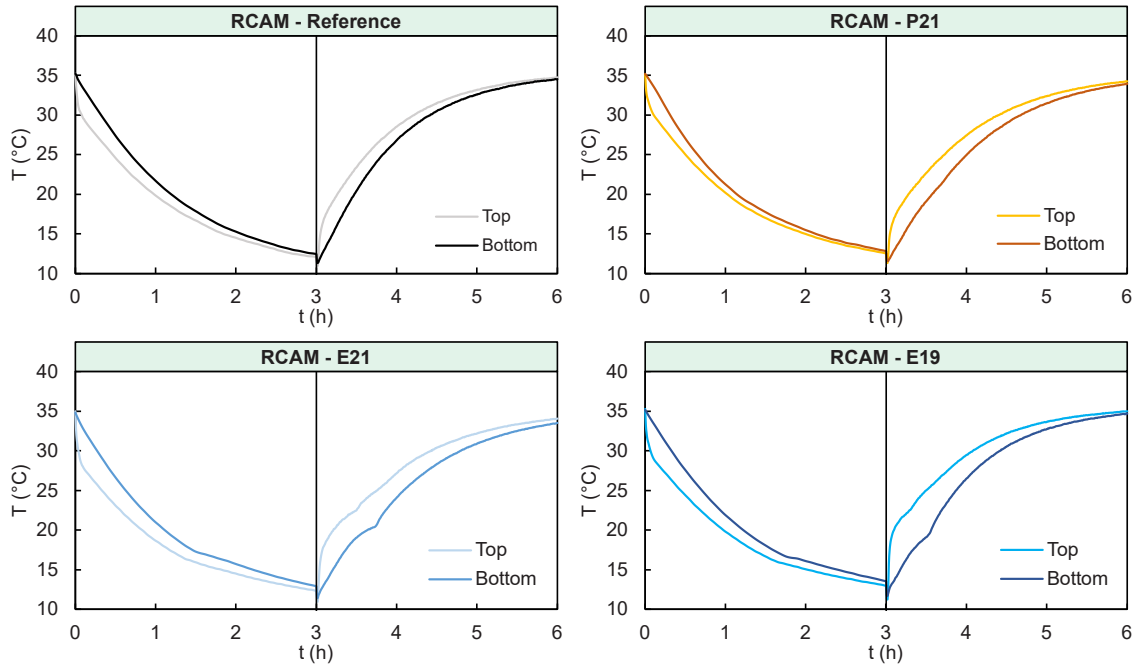


Fig. 13. Transient top and bottom temperature in the RCAM samples.

These observations are essential for designing materials tailored to specific thermal performance requirements, where the magnitude of thermal inertia and temperature stabilization are critical factors.

4. Conclusions and future work

This study incorporated three types of phase change materials (PCMs) into different mortars with varying porosities to investigate their impact on thermal, mechanical, and physical properties. A paraffin PCM and two ester PCMs were used in three types of mortars: natural aggregate mortar (NAM), recycled concrete aggregate mortar (RCAM), and lightweight aggregate mortar (LWAM). The following conclusions

were drawn based on the experimental results:

- Using a paraffin PCM with a 21 °C melting point and its impregnation in LWAM offered the best solution among all studied in terms of thermal properties. It showed the highest increase in specific heat capacity (42.92 %), maintaining a low conductivity (1.08 W/m·K).
- PCM incorporation in mortars leads to a tradeoff between the increase of the thermal conductivity and the improvement of the thermal inertia. In most cases, the rise in effusivity and specific heat outweighed the increase in conductivity, which is beneficial for reducing energy consumption in buildings. Overall, LWAM showed the best performance since it has the lowest conductivity and highest

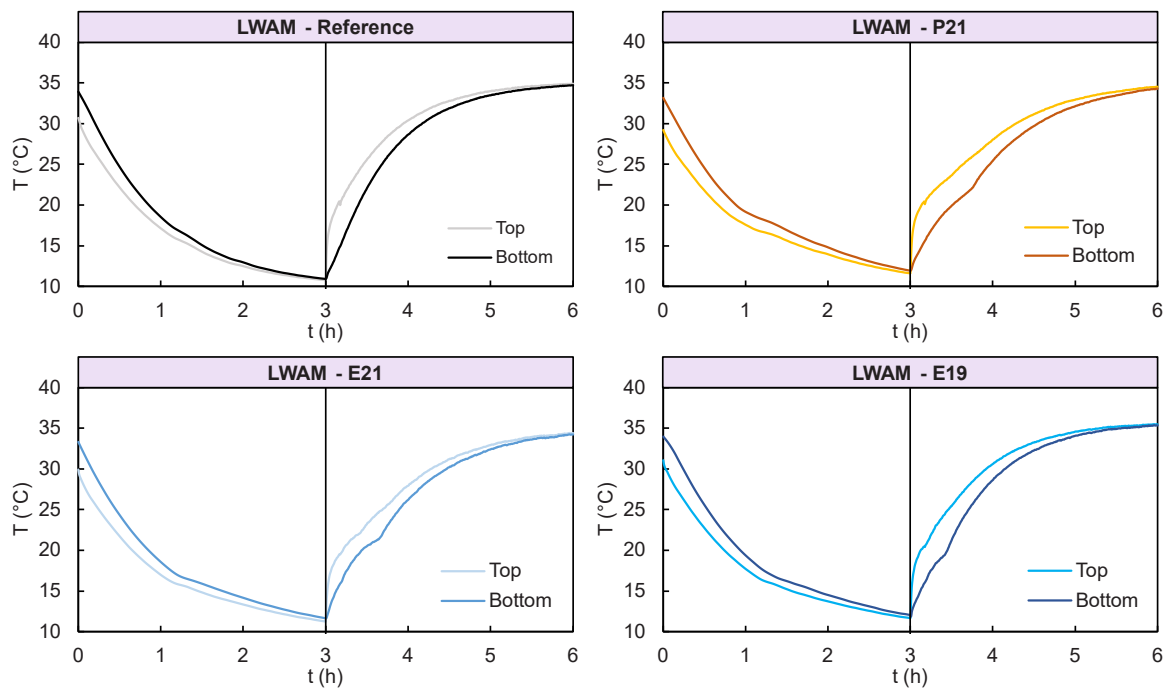


Fig. 14. Transient top and bottom temperature in the LWAM samples.

specific heat capacity after PCM incorporation. However, certain combinations of mortar-PCM might be especially interesting depending on the targeted specific heat and maximum admissible conductivity.

- The interaction between mortar porosity and PCM viscosity plays a decisive role in the incorporated mass of PCM. PCMs with low viscosity are better suited for mortars with lower porosity. In contrast, more viscous PCMs lead to greater incorporation for mortars with a higher porosity.
- PCM addition has a negligible impact on the mechanical properties of NAM and RCAM. In contrast, the mechanical performance of LWAM was improved, indicating that PCM incorporation slightly enhances the strength of porous mortars.
- The incorporation of PCM into mortars altered the transient thermal response. Mortars with PCM exhibit delayed temperature variations due to latent heat exchange during phase transitions. The specific combination of mortar type and PCM led to different thermal responses, indicating that the particular PCM must be carefully chosen depending on the application.

The experimental findings prove that PCM incorporation into mortars improves thermal performance while maintaining structural integrity, particularly in high-porosity materials like LWAM. This work provides a fundamental understanding of the mechanisms involved in PCM impregnation in cement-based composites. The analysis of the interaction between the aggregate and the PCM type contributes to the development of the incorporation of PCMs into precast elements, such as concrete blocks, which are widely used in construction. This approach ensures controlled impregnation while maintaining structural integrity and optimizing thermal efficiency, allowing for the design of thermal-efficient buildings targeted for specific climates.

To contribute to the deployment of this technology in building envelopes, the following future works are needed:

- Broadening the present work with other PCMs, mortar mixture formulations (altering, for example, the water-to-cement ratio), and incorporation times.

- Measuring the energy storage capacity and the dynamic thermal response of PCM-incorporated mortars applying temperature curves to actual size samples.
- Evaluating the long-term performance of PCM-impregnated mortars under repeated thermal cycling conditions to ensure their durability and stability. Such studies are essential to complement the current findings and validate the practical applicability.
- Advancing the thermal modeling techniques for thermal energy storage mortars. This is especially relevant to study the thermal performance of building envelopes or even whole building systems that incorporate these materials.
- Performing the life cycle and economic analysis of mortars with PCM to evaluate their practical application in sustainable construction.

CRediT authorship contribution statement

Lopez-Arias Marina: Writing – review & editing, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Alvarez-Rodriguez Matias:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Alonso-Martinez Mar:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Velay-Lizancos Mirian:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.conbuildmat.2025.140642](https://doi.org/10.1016/j.conbuildmat.2025.140642).

Data Availability

Data will be made available on request.

References

- [1] International Energy Agency, "Tracking Clean Energy Progress 2023", Paris, 2024. Accessed: Apr. 23, 2024. [Online]. Available: <https://www.iea.org/reports/tracking-clean-energy-progress-2023>
- [2] International Energy Agency, "Tracking Buildings 2021 – IEA", Paris, 2021. Accessed: Jan. 26, 2022. [Online]. Available: <https://www.iea.org/reports/buildings>
- [3] International Energy Agency et al., "Net Zero by 2050: A Roadmap for the Global Energy Sector", Paris, 2021. Accessed: Sep. 16, 2024. [Online]. Available: <https://trid.trb.org/view/1856381>
- [4] European Commission, "Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions. The European Green Deal", 2019, Accessed: Feb. 28, 2022. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1588580774040&uri=CELEX%3A52019DC0640>
- [5] European Commission, "Regulation (EU) 2021/1119 of the European Parliament and of the Council of 30 June 2021 establishing the framework for achieving climate neutrality and amending Regulations (EC) No 401/2009 and (EU) 2018/1799 ('European Climate Law')", *Official Journal of the European Union L243*, pp. 1–17, 2021, Accessed: Feb. 28, 2022. [Online]. Available: <https://eur-lex.europa.eu/legal-content/ES/TXT/?uri=CELEX:32021R1119>
- [6] United Nations Organization, "Transforming our world: the 2030 Agenda for Sustainable Development", 2016. Accessed: Sep. 16, 2024. [Online]. Available: <https://sdgs.un.org/2030agenda>
- [7] H. Jouhara, A. Żabnieńska-Góra, N. Khordehghah, D. Ahmad, T. Lipinski, Latent thermal energy storage technologies and applications: a review, *Int. J. Thermofluids* 5–6 (2020), <https://doi.org/10.1016/j.ijft.2020.100039>.
- [8] X. Jin, et al., A new heat transfer model of phase change material based on energy asymmetry, *Appl. Energy* 212 (2018) 1409–1416, <https://doi.org/10.1016/j.apenergy.2017.12.103>.
- [9] S.R.L. da Cunha, J.L.B. de Aguiar, Phase change materials and energy efficiency of buildings: a review of knowledge, *J. Energy Storage* 27 (2020) 101083, <https://doi.org/10.1016/j.est.2019.101083>.
- [10] K. Faraj, M. Khaled, J. Faraj, F. Hachem, C. Castelain, A review on phase change materials for thermal energy storage in buildings: Heating and hybrid applications, *J. Energy Storage* 33 (2021) 101913, <https://doi.org/10.1016/j.est.2020.101913>.
- [11] Á. De Gracia, L.F. Cabeza, Phase change materials and thermal energy storage for buildings, *Energy Build.* 103 (2015) 414–419, <https://doi.org/10.1016/j.enbuild.2015.06.007>.
- [12] U. Berardi, A.A. Gallardo, Properties of concretes enhanced with phase change materials for building applications, *Energy Build.* 199 (2019) 402–414, <https://doi.org/10.1016/j.enbuild.2019.07.014>.
- [13] A. El Majd, et al., Advancing PCM research in building efficiency: a comprehensive investigation into PCM selection and critical integration strategies, *J. Build. Eng.* 96 (2024) 110485, <https://doi.org/10.1016/j.job.2024.110485>.
- [14] H. Yang, et al., Cementitious composites integrated phase change materials for passive buildings: An overview, *Constr. Build. Mater.* 361 (2022) 129635, <https://doi.org/10.1016/j.conbuildmat.2022.129635>.
- [15] Z.I. Djamai, F. Salvatore, A. Si Larbi, G. Cai, M. El Mankibi, Multiphysics analysis of effects of encapsulated phase change materials (PCMs) in cement mortars, *Cem. Concr. Res* 119 (2019) 51–63, <https://doi.org/10.1016/j.cemconres.2019.02.002>.
- [16] X. Wang, W. Li, Z. Luo, K. Wang, S.P. Shah, A critical review on phase change materials (PCM) for sustainable and energy efficient building: design, characteristic, performance and application, *Energy Build.* 260 (2022) 111923, <https://doi.org/10.1016/j.enbuild.2022.111923>.
- [17] R. Wen, et al., Preparation and thermal properties of fatty acid/diatomite form-stable composite phase change material for thermal energy storage, *Sol. Energy Mater. Sol. Cells* 178 (2018) 273–279, <https://doi.org/10.1016/j.solmat.2018.01.032>.
- [18] J. Nam, S. Yang, B.Y. Yun, S. Kim, Evaluation of thermal/morphological performance of SSPCM based nanoclay: Influence of the interlayer microstructure of hydrophilic and hydrophobic, *Sol. Energy Mater. Sol. Cells* 235 (2022) 111479, <https://doi.org/10.1016/j.solmat.2021.111479>.
- [19] L. Bi, G. Long, C. Ma, Y. Xie, Mechanical properties and water absorption of steam-cured mortar containing phase change composites, *Constr. Build. Mater.* 248 (2020) 118707, <https://doi.org/10.1016/j.conbuildmat.2020.118707>.
- [20] G. Hekimoğlu, A. Sari, S. Arunachalam, H. Arslanoğlu, O. Gencel, Porous biochar/heptadecane composite phase change material with leak-proof, high thermal energy storage capacity and enhanced thermal conductivity, *Powder Technol.* 394 (Dec. 2021) 1017–1025, <https://doi.org/10.1016/j.powtec.2021.09.030>.
- [21] H.S. Esmaeili, Y. Farnam, J.E. Haddock, P.D. Zavattieri, W.J. Weiss, Numerical analysis of the freeze-thaw performance of cementitious composites that contain phase change material (PCM), *Mater. Des.* 145 (2018) 74–87, <https://doi.org/10.1016/j.matdes.2018.02.056>.
- [22] M. Aguayo, S. Das, C. Castro, N. Kabay, G. Sant, N. Neithalath, Porous inclusions as hosts for phase change materials in cementitious composites: Characterization, thermal performance, and analytical models, *Constr. Build. Mater.* 134 (2017) 574–584, <https://doi.org/10.1016/j.conbuildmat.2016.12.185>.
- [23] E. Mohseni, W. Tang, S. Wang, Development of thermal energy storage lightweight structural cementitious composites by means of macro-encapsulated PCM, *Constr. Build. Mater.* 225 (Nov. 2019) 182–195, <https://doi.org/10.1016/j.conbuildmat.2019.07.136>.
- [24] A. Marani, M.L. Nehdi, Integrating phase change materials in construction materials: Critical review, *Constr. Build. Mater.* 217 (2019) 36–49, <https://doi.org/10.1016/j.conbuildmat.2019.05.064>.
- [25] L.F. Cabeza, A. Castell, C. Barreneche, Á. De Gracia, A.I. Fernández, Materials used as PCM in thermal energy storage in buildings: a review, *Renew. Sustain. Energy Rev.* 15 (3) (2011) 1675–1695, <https://doi.org/10.1016/j.rser.2010.11.018>.
- [26] Z. Liu, et al., Comprehensive investigations on printability and thermal performance of cementitious material incorporated with PCM under various conditions, *Energy Convers. Manag.* 261 (2022) 115667, <https://doi.org/10.1016/j.enconman.2022.115667>.
- [27] L.F. Cabeza, et al., Behaviour of a concrete wall containing micro-encapsulated PCM after a decade of its construction, *Sol. Energy* 200 (2020) 108–113, <https://doi.org/10.1016/j.solener.2019.12.003>.
- [28] J. Shi, et al., Thermal and mechanical properties of thermal energy storage lightweight aggregate mortar incorporated with phase change material, *J. Energy Storage* 32 (2020) 101719, <https://doi.org/10.1016/j.est.2020.101719>.
- [29] B. Klemczak, et al., Utilization of waste foam concrete with MPCM as a substitution material for cement in mortars, *J. Build. Eng.* 90 (2024) 109284, <https://doi.org/10.1016/j.job.2024.109284>.
- [30] P. Suttaphakdee, N. Dulsang, N. Lorwanishpaisarn, P. Kasemsiri, P. Posi, P. Chindaprasit, Optimizing mix proportion and properties of lightweight concrete incorporated phase change material paraffin/recycled concrete block composite, *Constr. Build. Mater.* 127 (2016) 475–483, <https://doi.org/10.1016/j.conbuildmat.2016.10.037>.
- [31] J. Rostami, O. Khandel, R. Sedighardkani, A.R. Sahneh, S.A. Ghahari, Enhanced workability, durability, and thermal properties of cement-based composites with aerogel and paraffin coated recycled aggregates, *J. Clean. Prod.* 297 (2021) 126518, <https://doi.org/10.1016/j.jclepro.2021.126518>.
- [32] C. Xu, H. Zhang, G. Fang, Review on thermal conductivity improvement of phase change materials with enhanced additives for thermal energy storage, *J. Energy Storage* 51 (2022) 104568, <https://doi.org/10.1016/j.est.2022.104568>.
- [33] E. Malachanne, M. Jebli, F. Jamin, E. Garcia-Diaz, M.S. El Youssefi, A cohesive zone model for the characterization of adhesion between cement paste and aggregates, *Constr. Build. Mater.* 193 (2018) 64–71, <https://doi.org/10.1016/j.conbuildmat.2018.10.188>.
- [34] S. Drissi, A. Eddahhak, S. Caré, J. Neji, Thermal analysis by DSC of Phase Change Materials, study of the damage effect, *J. Build. Eng.* 1 (2015) 13–19, <https://doi.org/10.1016/j.job.2015.01.001>.
- [35] M.F. Junaed, Z. ur Rehman, N. Ijaz, R. Farooq, U. Khalid, Z. Ijaz, Performance evaluation of cement-based composites containing phase change materials from energy management and construction standpoints, *Constr. Build. Mater.* 416 (2024) 135108, <https://doi.org/10.1016/j.conbuildmat.2024.135108>.
- [36] A. El Majd, et al., Advancing PCM research in building efficiency: a comprehensive investigation into PCM selection and critical integration strategies, *J. Build. Eng.* 96 (2024) 110485, <https://doi.org/10.1016/j.job.2024.110485>.
- [37] M. Lopez-Arias, V. Francioso, M. Velay-Lizancos, High thermal inertia mortars: New method to incorporate phase change materials (PCMs) while enhancing strength and thermal design models, *Constr. Build. Mater.* 370 (2023) 130621, <https://doi.org/10.1016/j.conbuildmat.2023.130621>.
- [38] Saint Gobain, "Arlita® S: expanded clay lightweight aggregate (in Spanish)". Accessed: Aug. 01, 2024. [Online]. Available: <https://www.arlita.es/productos/arlit-a-s>
- [39] ASTM International, "ASTM C305-20 Standard Practice for Mechanical Mixing of Hydraulic Cement Pastes and Mortars of Plastic Consistency". Accessed: Aug. 01, 2024. [Online]. Available: <https://www.astm.org/c305-20.html>.
- [40] Rubitherm Technologies GmbH, "RT21HC Data Sheet". Accessed: Aug. 01, 2024. [Online]. Available: https://www.rubitherm.eu/media/products/datasheets/Techdata-RT21HC_EN_09102020.PDF.
- [41] Croda International Plc, "CrodaTherm 21". Accessed: Aug. 01, 2024. [Online]. Available: https://www.crodaindustrialspecialties.com/en-gb/product-finder/product/1798-crodatherm_1_21.
- [42] Croda International Plc, "CrodaTherm 19". Accessed: Aug. 01, 2024. [Online]. Available: https://www.crodaindustrialspecialties.com/en-gb/product-finder/product/1793-crodatherm_1_19.
- [43] C. Castellón, E. Günther, H. Mehling, S. Hiebler, L.F. Cabeza, Determination of the enthalpy of PCM as a function of temperature using a heat-flux DSC - A study of different measurement procedures and their accuracy, *Int. J. Energy Res* 32 (13) (Oct. 2008) 1258–1265, <https://doi.org/10.1002/er.1443>.

- [44] ASTM International, "ASTM Standard E793(2018), Standard Test Method for Enthalpies of Fusion and Crystallization by Differential Scanning Calorimetry", 2006. doi: 10.1520/E0793-06R18.
- [45] ASTM International, "ASTM D7984 Standard Test Method for Measurement of Thermal Effusivity of Fabrics Using a Modified Transient Plane Source (MTPS) Instrument". Accessed: Aug. 04, 2024. [Online]. Available: <https://www.astm.org/d7984-21.html>.
- [46] ASTM International, "ASTM C830-00(2023) Standard Test Methods for Apparent Porosity, Liquid Absorption, Apparent Specific Gravity, and Bulk Density of Refractory Shapes by Vacuum Pressure". Accessed: Aug. 01, 2024. [Online]. Available: <https://www.astm.org/c0830-00r23.html>.
- [47] C.A. Schneider, W.S. Rasband, K.W. Eliceiri, NIH Image to ImageJ: 25 years of image analysis, *Nat. Methods* 9 (7) (2012) 671–675, <https://doi.org/10.1038/nmeth.2089>.
- [48] C. Moro, V. Francioso, M. Lopez-Arias, M. Velay-Lizancos, Modification of self-cleaning activity on cement pastes containing nano-TiO₂ due to CO₂ curing, *Constr. Build. Mater.* 330 (2022) 127185, <https://doi.org/10.1016/j.conbuildmat.2022.127185>.
- [49] M.M. Velay Lizancos, Methods of producing construction elements, construction elements produced thereby, and structures produced therefrom, *WO2023205162A1* (2023).
- [50] Y. Liu, et al., Study on Complexity of Precast Concrete Components and Its Influence on Production Efficiency, *Adv. Civ. Eng.* 2022 (1) (2022) 9926547, <https://doi.org/10.1155/2022/9926547>.
- [51] ASTM International, "ASTM C348 Standard Test Method for Flexural Strength of Hydraulic-Cement Mortars". Accessed: Aug. 04, 2024. [Online]. Available: <https://www.astm.org/c0348-21.html>.
- [52] ASTM International, "ASTM C349 Standard Test Method for Compressive Strength of Hydraulic-Cement Mortars (Using Portions of Prisms Broken in Flexure)". Accessed: Aug. 04, 2024. [Online]. Available: <https://www.astm.org/c0349-18.html>.
- [53] International Organization for Standardization, "ISO 22007-2:2022 - Plastics — Determination of thermal conductivity and thermal diffusivity — Part 2: Transient plane heat source (hot disc) method". Accessed: Aug. 04, 2024. [Online]. Available: <https://www.iso.org/standard/81836.html>.
- [54] M. Pomianowski, P. Heiselberg, R.L. Jensen, R. Cheng, Y. Zhang, A new experimental method to determine specific heat capacity of inhomogeneous concrete material with incorporated microencapsulated-PCM, *Cem. Concr. Res* 55 (2014) 22–34, <https://doi.org/10.1016/j.cemconres.2013.09.012>.
- [55] L. Klimeš, et al., Computer modelling and experimental investigation of phase change hysteresis of PCMs: The state-of-the-art review, *Appl. Energy* 263 (2020) 114572, <https://doi.org/10.1016/j.apenergy.2020.114572>.
- [56] M.H. Zahir, S.A. Mohamed, R. Saidur, F.A. Al-Sulaiman, Supercooling of phase-change materials and the techniques used to mitigate the phenomenon, *Appl. Energy* 240 (2019) 793–817, <https://doi.org/10.1016/j.apenergy.2019.02.045>.
- [57] L. Liu, X. Zhang, X. Xu, Y. Zhao, S. Zhang, The research progress on phase change hysteresis affecting the thermal characteristics of PCMs: a review, *J. Mol. Liq.* 317 (2020) 113760, <https://doi.org/10.1016/j.molliq.2020.113760>.
- [58] K.P. Verian, W. Ashraf, Y. Cao, Properties of recycled concrete aggregate and their influence in new concrete production, *Resour. Conserv. Recycl.* 133 (2018) 30–49, <https://doi.org/10.1016/j.resconrec.2018.02.005>.
- [59] J. Xiao, W. Li, Z. Sun, D.A. Lange, S.P. Shah, Properties of interfacial transition zones in recycled aggregate concrete tested by nanoindentation, *Cem. Concr. Compos* 37 (1) (2013) 276–292, <https://doi.org/10.1016/j.cemconcomp.2013.01.006>.
- [60] D.C. Montgomery. *Design and Analysis of Experiments*, 8th ed., John Wiley & Sons Inc, New York, 2012.